# Linux Rapid-Fire DevOps Questions (Interview-Ready Answers)

**1. What is Linux?**

- Linux is an open-source operating system kernel used to manage system resources.
- It provides process management, memory management, and file systems.
- Example: Ubuntu, CentOS, and Amazon Linux use the Linux kernel.

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**2. What is a process?**

- A process is a running instance of a program in the operating system.
- Each process has a unique Process ID (PID).
- Example: running nginx or java creates processes.

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**3. What is a zombie process?**

- A zombie process is a completed process that still exists in the process table.
- It happens when the parent process has not collected the child's exit status.
- Example: seen in ps aux with state Z.

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**4. What is a daemon process?**

- A daemon is a background service running continuously in the system.
- It usually starts during system boot.
- Example: sshd, systemd, crond.

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**5. What is inode?**

- An inode is a data structure storing metadata of a file.
- It contains permissions, owner, and file location but not the file name.
- Example: ls -i displays inode numbers.

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**6. Hard link vs soft link?**

- A hard link directly references the inode of a file.
- A soft link is a symbolic pointer to a file path.
- Example: ln file1 file2 vs ln -s file1 link.

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**7. What is swap space?**

- Swap is disk space used when system RAM is full.
- It temporarily stores inactive memory pages.
- Example: checked using free -m.

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**8. What is load average?**

- Load average represents system CPU demand over time.
- It shows how many processes are waiting for CPU resources.
- Example: uptime displays 1, 5, and 15 minute load averages.

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**9. What is grep?**

- grep is a command used to search text patterns inside files.
- It supports regular expression pattern matching.
- Example: grep error app.log.

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**10. What is chmod?**

- chmod changes file permissions in Linux.
- Permissions include read, write, and execute for users.
- Example: chmod 755 script.sh.

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**11. What is chown?**

- chown changes file ownership in Linux.
- It assigns user and group ownership.
- Example: chown ubuntu:dev file.txt.

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**12. What is the top command?**

- top displays real-time system resource usage.
- It shows CPU, memory, and running processes.
- Example: used to monitor high CPU usage.

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**13. What is ps command?**

- ps displays running processes in the system.
- It provides process ID, CPU usage, and command details.
- Example: ps aux.

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**14. What is kill command?**

- kill sends signals to terminate or control processes.
- It allows stopping processes using signal numbers.
- Example: kill -9 PID.

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**15. What is crontab?**

- Crontab schedules automated jobs in Linux.
- It runs commands at specific time intervals.
- Example: daily backup using cron job.

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**16. What is SSH?**

- SSH is a secure protocol used for remote server access.
- It encrypts communication between client and server.
- Example: ssh ubuntu@server-ip.

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**17. What is netstat?**

- netstat shows active network connections and ports.
- It helps troubleshoot network issues.

- Example: netstat -tuln.

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**18. What is tar command?**

- tar is used to archive and compress files.
- It combines multiple files into a single archive.
- Example: tar -czvf backup.tar.gz.

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**19. What is PATH variable?**

- PATH is an environment variable listing executable directories.
- It allows commands to run without specifying full path.
- Example: /usr/bin, /usr/local/bin.

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**20. What is /etc directory?**

- /etc contains system configuration files.
- It stores settings for system services and applications.
- Example: /etc/nginx/nginx.conf.

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**21. What is the difference between kill and kill -9?**

- kill sends a termination signal allowing process cleanup.
- kill -9 forcefully terminates the process immediately.
- Example: used when process is stuck.

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**22. What is uptime command?**

- uptime shows how long the system has been running.
- It also displays system load averages.
- Example: checking server health.

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**23. What is df command?**

- df shows disk space usage of file systems.

- It helps monitor available storage.
- Example: df -h.

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**24. What is du command?**

- du shows disk usage of files and directories.
- It helps find large directories consuming space.
- Example: du -sh folder.

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**25. What is free command?**

- free displays system memory usage.
- It shows RAM and swap utilization.
- Example: free -m.

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**26. What is systemctl?**

- systemctl manages system services in Linux.
- It starts, stops, and checks service status.
- Example: systemctl restart nginx.

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**27. What is the difference between su and sudo?**

- su switches to another user account.
- sudo runs a command with elevated privileges.
- Example: sudo apt update.

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**28. What is a file descriptor?**

- A file descriptor is a numeric reference for open files.
- It allows processes to read and write files.
- Example: 0 stdin, 1 stdout, 2 stderr.

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**29. What is mount?**

- Mount attaches a file system to a directory.
- It allows accessing storage devices.
- Example: mounting EBS volume.

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**30. What is a shell?**

- A shell is a command-line interface for interacting with the OS.
- It interprets user commands and executes them.
- Example: Bash shell.

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